

WEEK:- 6

Topic: JavaScript Basics, DOM Manipulation, Validation, Events, and Interactive Web Applications

Objectives:

To understand the fundamentals of JavaScript, learn DOM manipulation techniques, implement form validation, handle events, and develop interactive web applications with dynamic functionality.

Learning Outcomes:

- Learned JavaScript syntax, variables, functions, and control structures.
- Understood DOM manipulation and dynamic content updates.
- Gained knowledge of form validation and event handling techniques.
- Developed skills in creating interactive and user-friendly web applications.
- Improved problem-solving and frontend development capabilities.

WORK SUMMARY:

During Week 6, I worked on JavaScript basics, DOM manipulation, validation, events, and interactive web applications. The training focused on understanding how JavaScript enhances the functionality and interactivity of web pages. I attended technical sessions and participated in hands-on exercises to learn the core concepts of client-side scripting and dynamic webpage behavior. Through practical examples and guided implementation activities, I explored how JavaScript can be used to create responsive and interactive user experiences.

Throughout the week, I gained practical experience in working with variables, functions, loops, conditional statements, and objects. I learned how to manipulate the Document Object Model (DOM) to dynamically update webpage content, modify elements, and respond to user actions. Special emphasis was placed on event handling techniques such as click events, keyboard interactions, and form submissions. I also implemented validation mechanisms to ensure accurate user input and improve application reliability.

The activities carried out during this week enhanced my understanding of frontend development and interactive web technologies. I learned how JavaScript works together with HTML and CSS to create modern web applications. Regular coding exercises, debugging sessions, and mentor feedback helped me improve my logical thinking and problem-solving skills.

Activities Performed:

- Implemented JavaScript programs involving variables, functions, validation, and event handling.
- Performed DOM manipulation and developed interactive web application features.